# Band-1 Hadamard order 14000 – Classical Paley-I (q=13999); IN SPK via Paley-I; Gap 2052 Still OPEN

H_14000 Paley-I PASS; Paley-II FAIL/SKIP (q=6999=3*2333 NOT prime; ===3 mod 4 needs ===1); Kronecker SKIP deferred note-only (4*3500; 3500===0 mod 4 order-admissible; factors 2^2*5^3*7; not_in_seeds / no H_3500; H_3500 classically via Paley-I q=3499 prime ===3 mod 4 (direct) ALSO ELIGIBLE – deferred note-only, NOT constructed, not on box); Miyamoto digression skipped; gap 2052 OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-10-07

**Document class.** Scientific computational / construction certificate (Band-1).

**Program.** Open-mathematics collapse list * Human+AI attack order * ZoraASI mission-control.

**Parents.** 2026-10-07_Band1_Hadamard_13968_PaleyI; 2026-10-07_Band1_Hadamard_13964_PaleyI; 2026-10-07_Band1_Hadamard_13932_PaleyI; 2026-10-07_Band1_Hadamard_13908_PaleyI; 2026-10-07_Band1_Hadamard_13904_PaleyI; 2026-10-07_Band1_Hadamard_13884_PaleyI; 2026-10-07_Band1_Hadamard_13880_PaleyI; 2026-10-07_Band1_Hadamard_13860_PaleyI; 2026-10-07_Band1_Hadamard_13832_PaleyI; 2026-10-07_Band1_Hadamard_13808_PaleyI; 2026-10-07_Band1_Hadamard_13800_PaleyI; 2026-10-07_Band1_Hadamard_13764_PaleyI; 2026-10-07_Band1_Hadamard_13760_PaleyI; 2026-10-07_Band1_Hadamard_13752_PaleyI; 2026-10-07_Band1_Hadamard_13724_PaleyI; 2026-10-07_Band1_Hadamard_13712_PaleyI; 2026-10-07_Band1_Hadamard_13692_PaleyI; 2026-10-07_Band1_Hadamard_13688_PaleyI; 2026-10-06_Band1_Hadamard_13680_PaleyI; 2026-10-06_Band1_Hadamard_13628_PaleyI; 2026-10-06_Band1_Hadamard_13620_PaleyI; 2026-10-06_Band1_Hadamard_13592_PaleyI; 2026-10-06_Band1_Hadamard_13568_PaleyI; 2026-10-06_Band1_Hadamard_13524_PaleyI; 2026-10-06_Band1_Hadamard_13500_PaleyI; 2026-10-06_Band1_Hadamard_13488_PaleyI; 2026-10-06_Band1_Hadamard_13464_PaleyI; 2026-10-06_Band1_Hadamard_13452_PaleyI; 2026-10-06_Band1_Hadamard_13412_PaleyI; 2026-10-06_Band1_Hadamard_13400_PaleyI; 2026-10-06_Band1_Hadamard_13368_PaleyI; 2026-10-06_Band1_Hadamard_13340_PaleyI; 2026-10-06_Band1_Hadamard_13332_PaleyI; 2026-10-06_Band1_Hadamard_13328_PaleyI; 2026-10-06_Band1_Hadamard_13292_PaleyI; 2026-10-06_Band1_Hadamard_13268_PaleyI; 2026-10-06_Band1_Hadamard_13260_PaleyI; 2026-10-06_Band1_Hadamard_13220_PaleyI; 2026-10-06_Band1_Hadamard_13188_PaleyI; 2026-10-06_Band1_Hadamard_13184_PaleyI; 2026-10-06_Band1_Hadamard_13172_PaleyI; 2026-10-05_Band1_Hadamard_13164_PaleyI; 2026-10-05_Band1_Hadamard_13160_PaleyI; 2026-10-05_Band1_Hadamard_13152_PaleyI;

2026-10-05_Band1_Hadamard_13148_PaleyI; 2026-10-05_Band1_Hadamard_13128_PaleyI;
2026-10-05_Band1_Hadamard_13104_PaleyI; 2026-10-05_Band1_Hadamard_13100_PaleyI;
2026-10-05_Band1_Hadamard_13064_PaleyI; 2026-10-05_Band1_Hadamard_13044_PaleyI;
2026-10-05_Band1_Hadamard_13008_PaleyI; 2026-10-05_Band1_Hadamard_13004_PaleyI;
2026-10-05_Band1_Hadamard_12984_PaleyI; 2026-10-05_Band1_Hadamard_12980_PaleyI;
2026-10-05_Band1_Hadamard_12968_PaleyI; 2026-10-05_Band1_Hadamard_12960_PaleyI;
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2026-10-04_Band1_Hadamard_12672_PaleyI; 2026-10-04_Band1_Hadamard_12660_PaleyI;
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2026-10-03_Band1_Hadamard_12240_PaleyI; 2026-10-03_Band1_Hadamard_12228_PaleyI;
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2026-10-03_Band1_Hadamard_12072_PaleyI; 2026-10-03_Band1_Hadamard_12044_PaleyI;
2026-10-03_Band1_Hadamard_12012_PaleyI; 2026-10-03_Band1_Hadamard_12008_PaleyI;
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2026-10-02_Band1_Hadamard_11904_PaleyI; 2026-10-02_Band1_Hadamard_11888_PaleyI;
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2026-10-02_Band1_Hadamard_11840_PaleyI; 2026-10-02_Band1_Hadamard_11832_PaleyI;
2026-10-02_Band1_Hadamard_11828_PaleyI; 2026-10-02_Band1_Hadamard_11808_PaleyI;
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2026-09-30_Band1_Hadamard_11520_PaleyI; 2026-09-30_Band1_Hadamard_11504_PaleyI;
2026-09-30_Band1_Hadamard_11492_PaleyI; 2026-09-30_Band1_Hadamard_11484_PaleyI;
2026-09-30_Band1_Hadamard_11472_PaleyI; 2026-09-30_Band1_Hadamard_11468_PaleyI;
2026-09-30_Band1_Hadamard_11448_PaleyI; 2026-09-30_Band1_Hadamard_11444_PaleyI;

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2026-09-30_Band1_Hadamard_11400_PaleyI; 2026-09-30_Band1_Hadamard_11384_PaleyI;
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2026-09-30_Band1_Hadamard_11280_PaleyI; 2026-09-30_Band1_Hadamard_11252_PaleyI;
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2026-09-29_Band1_Hadamard_11172_PaleyI; 2026-09-29_Band1_Hadamard_11160_PaleyI;
2026-09-29_Band1_Hadamard_11132_PaleyI; 2026-09-29_Band1_Hadamard_11120_PaleyI;
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2026-09-29_Band1_Hadamard_10980_PaleyI; 2026-09-29_Band1_Hadamard_10940_PaleyI;
2026-09-29_Band1_Hadamard_10892_PaleyI; 2026-09-29_Band1_Hadamard_10884_PaleyI;
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2026-09-28_Band1_Hadamard_10400_PaleyI; 2026-09-24_Band1_Hadamard_10392_PaleyI;
2026-09-24_Band1_Hadamard_10344_PaleyI; 2026-09-24_Band1_Hadamard_10332_PaleyI;
2026-09-24_Band1_Hadamard_10304_PaleyI; 2026-09-24_Band1_Hadamard_10272_PaleyI;
2026-09-23_Band1_Hadamard_10268_PaleyI; 2026-09-23_Band1_Hadamard_10224_PaleyI;
2026-09-23_Band1_Hadamard_10212_PaleyI; 2026-09-23_Band1_Hadamard_10164_PaleyI;
2026-09-23_Band1_Hadamard_10160_PaleyI; 2026-09-23_Band1_Hadamard_10152_PaleyI;
2026-09-23_Band1_Hadamard_10140_PaleyI; 2026-09-23_Band1_Hadamard_10112_PaleyI;
2026-09-23_Band1_Hadamard_10104_PaleyI; 2026-09-23_Band1_Hadamard_10100_PaleyI;
2026-09-23_Band1_Hadamard_10068_PaleyI.

**Lab.** `/workspace/experiments/band1/hadamard_14000_paleyI/`.

**Cutoff.** 2026-10-08T03:08:29Z (~11:08PM ET).

**Honesty stamp (non-negotiable).**

This memo claims **finite construction of order 14000 only (verified max|HHˆT-nI|=0), via classical Paley-I q=13999 (prime ===3 mod 4). 14000 IN SPK via Paley-I.** Honest skip: Miyamoto digression this wave (prefer classical SPK). Classical SPK for 14000: Paley-I **PASS**

($q = 13999$ prime $\equiv 3 \pmod 4$); Paley-II **FAIL/SKIP** (q=6999=3*2333 NOT prime; ===3 mod 4 needs ===1); Kronecker **SKIP deferred note-only** (note $14000 = 4 \cdot 3500$; $3500 \equiv 0 \pmod 4$ order-admissible; factors $2^2 \cdot 5^3 \cdot 7$; not_in_seeds / no H_3500; H_3500 classically via Paley-I q=3499 prime ===3 mod 4 (direct) ALSO ELIGIBLE – deferred note-only, NOT constructed, not on box). **14000** folds into SPK **only** via classical Paley-I (this run). It **does not** claim the full Hadamard conjecture. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims. Gap **2052** remains **OPEN**. `clay_closed = false`. Prior SPK **13524+13500+13488+13464+13452+13412+13400+13368+13340+13332+13328+13292+13268+132** retained; Miyamoto-only **8948/8644/8516/8452/7988/...** excluded.

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## Abstract

Finite construction of order **14000**.

- $H_{14000}$ = Paley-I ($q = 13999$) – classical SPK
- Paley-II for 14000: **FAIL/SKIP** ($q = 6999 = 3 \cdot 2333$ NOT prime; $\equiv 3 \pmod 4$ needs $\equiv 1$)
- Kronecker for 14000: **SKIP deferred note-only** ($4 \cdot 3500$; $3500 \equiv 0 \pmod 4$ order-admissible; factors $2^2 \cdot 5^3 \cdot 7$; not_in_seeds / no H_3500; H_3500 classically via Paley-I q=3499 prime ===3 mod 4 (direct) ALSO ELIGIBLE – deferred note-only, NOT constructed, not on box)
- Skip: Miyamoto digression this wave

**SPK: IN SPK via Paley-I**.
**Gap 2052:** OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

```
python3 verify_hadamard_14000.py
```

**Results.** overall_pass=True; units 635/635; row blocks 110/110 (block size 128); wall 2805.8945s; constructed max|HHˆT-nI|=0.

Cutoff: 2026-10-08T03:08:29Z (~11:08PM ET)

## Arithmetic

- Target $n = 14000 = q + 1$ with $q = 13999$ prime $\equiv 3 \pmod 4$ — Paley-I eligible.
- Paley-II: $q = 6999 = 3 \cdot 2333$ NOT prime; $\equiv 3 \pmod 4$ needs $\equiv 1$ — FAIL/SKIP.
- Kronecker: $14000 = 4 \cdot 3500$; $3500 \equiv 0 \pmod 4$ order-admissible; factors $2^2 \cdot 5^3 \cdot 7$; not_in_seeds / no H_3500; H_3500 classically via Paley-I q=3499 prime ===3 mod 4 (direct) ALSO ELIGIBLE – deferred note-only, NOT constructed, not on box — SKIP deferred note-only.
- Miyamoto digression skipped this wave (prefer classical SPK).

## Constructions (verified)

| Object | Method | max\|HHˆT-nI\| | Wall (s) |
|---|---|---|---|
| H_14000 Paley-I | Paley-I q=13999 | 0 | 2767.9865 |
| H_14000 Paley-II | FAIL/SKIP (q=6999=3*2333 NOT prime; ===3 mod 4 needs ===1) | – | – |
| H_14000 Kronecker | SKIP deferred note-only (*4**3500; 3500===0 mod 4 order-admissible; factors 2ˆ2*5ˆ3*7; not_in_seeds / no H_3500; H_3500 classically via Paley-I q=3499 prime ===3 mod 4 (direct) ALSO ELIGIBLE – deferred* note-only, NOT constructed, not on box) | – | – |

## SPK closure note

Seeds include prior SPK through **11528** lineage (**11528**; **11520**; **11504**; **11492**; **11484**; **11472**; **11468**; **11448**; **11444**; **11424**; **11412**; **11400**; **11384**; **11352**; **11312**; **11300**; **11288**; **11280**; **11252**; **11244**; **11240**; **11172**; **11160**; **11132**; **11120**; **11088**; **11084**; **11072**; **11060**; **11048**; **11028**; **11004**; **10988**; **10980**; **10940**; **10904**; **10892**; **10884**; **10860**; **10848**; **10832**; **10800**; **10772**; **10740**; **10724**; **10712**; **10692**; **10688**; **10668**; **10652**; **10640**; **10632**; **10628**; **10608**; **10568**; **10560**; **10532**; **10500**; **10488**; **10464**; **10460**; **10428**; **10400**; **10392**; **10344**; **10332**; **10304**; **10272**; **10268**; **10260**; **10248**; **10244**; **10224**; **10212**; **10164**; **10160**; **10152**; **10140**; **10112**; **10104**; **10100**; **10068**; **10040**; **10008**; **9968**; **9932**; **9924**; **9908**; **9888**; **9884**; **9872**; **9860**; **9852**; **9840**; **9812**; **9804**; **9792**; **9788**; **9768**; **9744**; **9740**; **9720**; **9680**; **9644**; **9632**; **9624**; **9620**; **9588**; **9552**; **9548**; **9540**; **9512**; **9492**; **9480**; **9468**; **9420**; **9404**; **9392**; **9372**; **9344**;

**9324**; **9320**; **9312**; **9284**; **9240**; **9228**; **9204**; **9200**; **9188**; **9152**; **9104**; **9128**; **9124**; **9092**; **2220+8884**; **2236**; **2160**; **2140+8564**; . . . ; **NOT** 8948/8644/8516/. . . ) + verified **10464** (+ prior **10392+10344+10332+10304+10272+10268+10224+10212+10164+10160+10152+10140+10112+101**
Next constructive gap **2052** (OPEN).

| Quantity | Value |
|---|---|
| 12488 in SPK | **True** (Paley-I q=13999) |
| Gap 2052 | **OPEN** |
| clay_closed | false |
| GATE-04 | LOCKED |

## Honesty

- clay_closed = false
- GATE-04 LOCKED
- Hadamard conjecture OPEN
- Gap 2052 OPEN (method triage / non-finding only)
- No RH / P!=NP / YM / Frankl / NS / AC solved claims
- SPK membership only via classical Paley-I/II/Kronecker (not Miyamoto alone)

## Next optional

q2=6961->n=13924 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=6961 prime ===1 mod 4; Paley-I q=13923=3^2*71317 *NOT prime -> FAIL; Kronecker 13924=4*3481; 3481=59^2 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3481); 13924=2^2*59^2 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=6977->n=13956 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=6977 prime ===1 mod 4; Paley-I q=13955=5*2791* NOT prime -> FAIL; Kronecker 13956=4*3489; 3489=3*1163 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3489); 13956=2^2*3*1163 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=6997->n=13996 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=6997 prime ===1 mod 4; Paley-I q=13995=3^2*5*311 NOT prime -> FAIL; Kronecker 13996=4*3499; 3499 odd prime (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3499); 13996=2^2*3499 ===12 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=7001->n=14004 Paley-II DIRECT (FIRST n===0 mod 4 after 14000 with any classical Paley route; q2=7001 prime ===1 mod 4; Paley-I q=14003=11*1967 NOT prime -> FAIL; Kronecker 14004=4*3501; 3501=3*1389 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3501); 14004=2^2*3*1389 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | next Paley-I DIRECT: q=14011->n=14012 (prime ===3 mod 4; 14009 prime ===1 mod 4 skipped; no prime ===3 mod 4 in 14000..14010; Paley-II q=7005=3*5*467 NOT prime (===1 mod 4 but composite) -> FAIL/SKIP; Kronecker 14012=4*3503; 3503=31*113 odd (===3 mod 4) -> NOT Hadamard-admissible ->

FAIL/SKIP (do NOT invent H_3503); 14012=2^2*31*113 ===12 mod 16 = 13964-like odd-m shape) (or gap 2052 / Track B)